Business Problem

Context:

Earthquake risk is a critical challenge for insurance companies, government agencies, and infrastructure planners. Raw seismic telemetry data from the USGS API is highly detailed but too granular and inconsistent for direct business use. It arrives as individual event records in JSON format, often containing duplicates, missing values, or schema mismatches.

Challenge:

Stakeholders need aggregated, standardized insights to make decisions, but they face several obstacles:

- Granularity: Individual earthquake events don't reveal long- term seismic trends.
- Data Quality: Raw feeds contain duplicates, nulls, and inconsistent severity measures.
- Business Readiness: Actuaries and analysts require datasets that translate technical signals into risk categories, geospatial bins, and severity scores.

Problem Statement:

"How can we transform raw, noisy earthquake telemetry into standardized, business- ready datasets that actuaries, planners, and analysts can use to model seismic risk, price insurance premiums, and prepare disaster response strategies?"

Why It Matters:

- Insurance & Reinsurance: Accurate premium pricing in earthquake- prone zones.
- Urban Planning: Identifying high- risk regions for infrastructure development.
- Disaster Preparedness: Supporting government agencies in allocating resources to vulnerable areas.